

HDOC DAY – 29

ALGORITHM:

1. Start.
2. Display the menu:
 - Count even, odd, positive, negative, and zero elements.
 - Search for an element using linear search.
 - Merge two arrays.
 - Exit.
3. Read the user's choice.
4. Use a switch statement:
 - Choice 1: Read an array and count even, odd, positive, negative, and zero elements.
 - Choice 2: Read an array and the element to search. Use linear search to find it.
 - Choice 3: Read two arrays and copy their elements into a third merged array.
 - Choice 4: Exit.
 - Otherwise, display "Invalid choice".
5. Repeat the menu until the user selects Exit.
6. Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int choice;
```

```
    int a[100], b[100], c[200];
```

```
    int n, n1, n2;
```

```
int i, key, found;

int even, odd, positive, negative, zero;

do
{
    printf("\n----- ARRAY OPERATIONS ----- \n");

    printf("1. Count even, odd, positive, negative and zero elements\n");

    printf("2. Linear search\n");

    printf("3. Merge two arrays\n");

    printf("4. Exit\n");

    printf("Enter your choice: ");

    scanf("%d", &choice);

    switch (choice)
    {
        case 1:

            printf("Enter the number of elements: ");

            scanf("%d", &n);

            if (n <= 0 || n > 100)
            {
                printf("Invalid array size.\n");

                break;
            }

            printf("Enter %d elements:\n", n);
```

```
for (i = 0; i < n; i++)  
{  
    scanf("%d", &a[i]);  
}
```

```
even = 0;
```

```
odd = 0;
```

```
positive = 0;
```

```
negative = 0;
```

```
zero = 0;
```

```
for (i = 0; i < n; i++)  
{  
    /* Check even or odd */  
    if (a[i] % 2 == 0)  
        even++;  
    else  
        odd++;  
  
    /* Check positive, negative or zero */  
    if (a[i] > 0)  
        positive++;  
    else if (a[i] < 0)  
        negative++;  
    else  
        zero++;  
}
```

```
printf("\nEven elements = %d\n", even);  
printf("Odd elements = %d\n", odd);  
printf("Positive elements = %d\n", positive);  
printf("Negative elements = %d\n", negative);  
printf("Zero elements = %d\n", zero);
```

```
break;
```

case 2:

```
printf("Enter the number of elements: ");  
scanf("%d", &n);
```

```
if (n <= 0 || n > 100)
```

```
{  
    printf("Invalid array size.\n");  
    break;  
}
```

```
printf("Enter %d elements:\n", n);
```

```
for (i = 0; i < n; i++)
```

```
{  
    scanf("%d", &a[i]);  
}
```

```
printf("Enter the element to search: ");
```

```
scanf("%d", &key);
```

```
found = 0;
```

```
/* Linear search */
```

```
for (i = 0; i < n; i++)
```

```
{
```

```
    if (a[i] == key)
```

```
    {
```

```
        printf("%d found at position %d.\n", key, i + 1);
```

```
        found = 1;
```

```
        break;
```

```
    }
```

```
}
```

```
if (found == 0)
```

```
{
```

```
    printf("%d not found in the array.\n", key);
```

```
}
```

```
break;
```

```
case 3:
```

```
    printf("Enter the number of elements in first array: ");
```

```
    scanf("%d", &n1);
```

```
    printf("Enter the number of elements in second array: ");
```

```
    scanf("%d", &n2);
```

```
    if (n1 <= 0 || n1 > 100 || n2 <= 0 || n2 > 100)
```

```
{
    printf("Invalid array size.\n");
    break;
}

printf("Enter %d elements of first array:\n", n1);

for (i = 0; i < n1; i++)
{
    scanf("%d", &a[i]);
}

printf("Enter %d elements of second array:\n", n2);

for (i = 0; i < n2; i++)
{
    scanf("%d", &b[i]);
}

/* Copy first array */
for (i = 0; i < n1; i++)
{
    c[i] = a[i];
}

/* Copy second array after first array */
for (i = 0; i < n2; i++)
{
```

```

        c[n1 + i] = b[i];
    }

    printf("Merged array:\n");

    for (i = 0; i < n1 + n2; i++)
    {
        printf("%d ", c[i]);
    }

    printf("\n");
    break;

case 4:

    printf("Exiting program...\n");
    break;

default:

    printf("Invalid choice. Please enter 1 to 4.\n");
}

} while (choice != 4);

return 0;
}

```

TEST CASE TABLE:

TEST	CHOICE	INPUT	EXPECTED OUTPUT
------	--------	-------	-----------------

1	1	2 -3 0 7 -4 8	Even = 4, Odd = 2, Positive = 3, Negative = 2, Zero = 1
2	2	Array: 10 20 30 40 50, Search: 30	30 found at position 3
3	2	Array: 10 20 30 40 50, Search: 60	60 not found in the array
4	3	A = 1 2 3, B = 4 5 6	Merged = 1 2 3 4 5 6
5	4	--	Program exits
6	4	--	Invalid choice

```
(kali㉿kali)-[~]
$ vi h29.c

(kali㉿kali)-[~]
$ gcc h29.c -o h29

(kali㉿kali)-[~]
$ ./h29

——— ARRAY OPERATIONS ———
1. Count even, odd, positive, negative and zero elements
2. Linear search
3. Merge two arrays
4. Exit
Enter your choice: 1
Enter the number of elements: 6
Enter 6 elements:
2 -3 0 -4 8
7

Even elements = 4
Odd elements = 2
Positive elements = 3
Negative elements = 2
Zero elements = 1
```